

module 2

day 7

Absolutely 🙌 Since this is for a **final exam**, I'll turn the huge EIGRP material into **easy, high-value exam notes**. I'll focus on what you should **remember, understand, and calculate**, then give you **2 easy + 2 hard examples**.

☒ EIGRP Final Exam – Important Notes

1. What is EIGRP?

EIGRP = Enhanced Interior Gateway Routing Protocol

It is a Cisco routing protocol used by routers to exchange routes and find the **best path** to destination networks.

Remember:



EIGRP = Advanced Distance Vector + DUAL + Fast Convergence

Originally, EIGRP was **Cisco proprietary**.



2. ☆ Main Features of EIGRP

Feature	Easy meaning
DUAL	Finds loop-free best and backup paths
Neighbor Table	Stores directly connected EIGRP neighbors
Topology Table	Stores learned routes from neighbors
Routing Table	Stores the best routes used for forwarding
RTP	Provides reliable delivery of EIGRP packets
Partial Updates	Sends only changed information
Bounded Updates	Sends updates only to affected routers
Load Balancing	Can use multiple paths
IPv4/IPv6	Supports multiple network protocols through PDMs

☆ Exam sentence



EIGRP provides **fast convergence, loop-free routing, efficient updates, and load balancing.**



3. ☒ EIGRP Tables

There are **3 important tables** to remember:

① Neighbor Table

Contains information about **directly connected EIGRP neighbors**.

Example:

R1 — R2

Copy

R1's neighbor table contains R2.



② Topology Table

Contains **all routes learned from EIGRP neighbors**.

It may contain:

- Best route
- Backup route
- Other possible routes



③ Routing Table

Contains the **best routes selected by DUAL**.

In:

```
show ip route
```

[Copy](#)

EIGRP routes are marked with:

```
D
```

[Copy](#)

☆ Easy memory



Neighbor → who is my neighbor?

Topology → what routes do I know?

Routing → which route will I use?



4. ☆ DUAL

DUAL = Diffusing Update Algorithm

This is one of the **most important EIGRP exam topics**.

DUAL provides:

- Loop-free paths
- Fast convergence

- Backup paths
- Best-path selection

Example

Suppose:

```
R2
 / \
R1  R3
 \ /
Network
```

Copy

R1 might use R2 as the primary path.

If R2 fails, EIGRP can quickly use R3 **if R3 is a Feasible Successor**.



5. ☆ Successor vs Feasible Successor

This is **very important**.

Successor

The **best path** to a destination.

Feasible Successor

A **backup path** that is already proven to be loop-free.

Easy memory:



Successor = Main road 🚧

Feasible Successor = Backup road □



6. ☆ FD and RD

Feasible Distance (FD)

The **best total metric** from the local router to the destination.

Reported Distance (RD)

The metric that a **neighbor reports** for reaching the destination.



7. ☆ Feasibility Condition

A neighbor becomes a **Feasible Successor** when:

[
RD < FD
]

In words:



Neighbor's RD must be less than my current FD.

Example

Local router:

FD = 3000

Copy

Neighbor reports:

RD = 2000

Copy

Check:

$2000 < 3000$

Copy

☐ True

Therefore, the neighbor can be a **Feasible Successor**.





8. EIGRP Packet Types ☆☆☆

There are **5 packet types** you should memorize.

Packet	Purpose	Reliable?
Hello	Discover/maintain neighbors	× No
Update	Send routing information	☐ Yes
ACK	Acknowledge reliable packets	× No
Query	Ask neighbors for a route	☐ Yes
Reply	Answer a Query	☐ Yes

Memory trick

H-U-A-Q-R



Hello → Update → ACK → Query → Reply



9. Hello Packet

Hello packets are used to:

- Discover neighbors
- Maintain neighbor relationships

IPv4 EIGRP multicast:

```
224.0.0.10
```

 Copy

IPv6:

```
FF02:: A
```

 Copy

Standard timers

```
Hello = 5 seconds  
Hold = 15 seconds
```

 Copy

Remember:

 5 / 15

If the hold timer expires, the neighbor is considered **down**.



10. ☆ EIGRP Reliable Transport Protocol

RTP = Reliable Transport Protocol

Don't confuse this with TCP.

EIGRP has its **own reliable transport mechanism**.

Reliable packets

- Update
- Query
- Reply

These require ACKs.

Unreliable packets

- Hello
- ACK

Easy memory:



UQR = Reliable
HA = Unreliable

11. EIGRP Multicast Address

For IPv4:

224.0.0.10

Copy

Very common exam question.

Also:

IP Protocol = 88

Copy

☆ Memorize



EIGRP = Protocol 88 = 224.0.0.10



12. Partial and Bounded Updates

This is another important difference from RIP.

RIP

Normally sends the entire routing table periodically.

EIGRP

Does **not** send the entire routing table periodically.

Instead:

Partial

Only the **changed route information** is sent.

Bounded

The update is sent only to **affected routers**.

Easy example

If R1 learns that:

```
192.168.10.0/24
```

[Copy](#)

has changed, it doesn't resend every route it knows.

It sends the relevant change.

☐ Saves bandwidth.



13. EIGRP AS Number

Configuration:

```
router eigrp 100
```

Copy

Here:

```
100
```

Copy

is the EIGRP AS number.

Important

Routers must use the **same EIGRP AS number** to form an adjacency.

Example:

```
R1 → router eigrp 100  
R2 → router eigrp 100
```

Copy

☐ Can form neighbor relationship.

But:

```
R1 → router eigrp 100  
R2 → router eigrp 200
```

[Copy](#)

× No adjacency.

⚠ Don't confuse

EIGRP AS number is **not the same thing as an Internet BGP AS number**.



14. EIGRP Router ID

Router ID must be:

- Unique
- Non-zero
- Not `255.255.255.255`

Priority:

1 Manually configured

```
eigrp router-id 1.1.1.1
```

[Copy](#)

2 Highest loopback IP

3 Highest active physical interface IP

Memory:



Manual → Loopback → Physical



15. EIGRP Network Command

Example:

```
router eigrp 100  
network 192.168.10.0
```

Copy

This enables EIGRP on matching interfaces.

You can also use a wildcard mask:

```
network 192.168.10.0 0.0.0.255
```

Copy



16. ☆ Wildcard Mask

Wildcard mask is the opposite of the subnet mask.

Formula:

[
Wildcard = 255.255.255.255 - Subnet\ Mask
]

Example

Subnet:

255.255.255.0

Copy

Wildcard:

0.0.0.255

Copy

Another:

255.255.255.252

Copy

Wildcard:

☆ Memorize common ones

Prefix	Subnet Mask	Wildcard
/24	255.255.255.0	0.0.0.255
/25	255.255.255.128	0.0.0.127
/26	255.255.255.192	0.0.0.63
/27	255.255.255.224	0.0.0.31
/28	255.255.255.240	0.0.0.15
/29	255.255.255.248	0.0.0.7
/30	255.255.255.252	0.0.0.3



17. Passive Interface

Command:

```
passive-interface gigabitEthernet 0/1
```

 Copy

It prevents EIGRP **Hello packets** from being sent through that interface.

Therefore, no EIGRP neighbor adjacency forms there.

Common configuration

```
passive-interface default  
no passive-interface g0/0
```

Copy

Meaning:



Make everything passive, then allow EIGRP on G0/0.



18. ☆ EIGRP Metric

By default, EIGRP mainly uses:

1. Bandwidth
2. Delay

Default K-values:

```
K1 = 1  
K2 = 0  
K3 = 1
```

Copy

$K4 = 0$

So by default:



Bandwidth + Delay

are used.

NOT normally used by default:

- Load ×
- Reliability ×



19. Bandwidth

EIGRP uses the **lowest bandwidth along the path**.

Example:

R1 — 100 Mbps — R2 — 10 Mbps — R3

Copy

EIGRP considers:

Minimum bandwidth = 10 Mbps

Copy

Important

The `bandwidth` command changes the **routing metric value**, not the actual physical speed.

Example:

```
interface g0/0  
bandwidth 10000
```

Copy

This does **not physically turn the interface into 10 Mbps**.



20. Delay

EIGRP adds the delay of interfaces along the path.

Example:

```
R1 → 10 μs  
R2 → 20 μs  
R3 → 10 μs
```

Copy

Total:

$$\begin{aligned} &[\\ &10 + 20 + 10 = 40 \mu s \\ &] \end{aligned}$$



21. ☆ EIGRP Metric Formula

With default K-values:

$$\begin{aligned} &[\\ &\text{Metric} = \\ &\left[\frac{10,000,000}{\text{Minimum Bandwidth}} \right. \\ &\quad \left. \bullet \frac{\text{Total Delay}}{10} \right] \\ &\quad \times 256 \\ &] \end{aligned}$$

Bandwidth is in **kbps**.



● EASY EXAMPLE 1 – EIGRP Concept

Question

R1 and R2 are connected using EIGRP.

R1 uses:

```
router eigrp 100
```

[Copy](#)

R2 uses:

```
router eigrp 100
```

[Copy](#)

Will they form an EIGRP neighbor relationship?

Answer

☐ **Yes**, assuming:

- EIGRP is enabled on the correct interfaces
- Interfaces are up
- IP addresses are compatible
- K-values match
- They can reach each other

Why?

Both routers use the same EIGRP AS:

100 = 100

Copy



● EASY EXAMPLE 2 – Feasible Successor

R1 has:

Current FD = 5000

Copy

A neighbor reports:

RD = 3000

Copy

Check:

[
RD < FD
]

[
3000 < 5000
]

☐ True.

Therefore:



The neighbor can become a **Feasible Successor**.



HARD EXAMPLE 1 – Metric Calculation

Suppose a route has:

Minimum bandwidth = 1000 kbps

Total delay = 20,000 μ s

Copy

Formula:

$$\left[\begin{aligned} &\text{Metric} = \\ &\frac{10,000,000}{1000} \\ &\bullet \frac{20,000}{10} \\ &\right] \times 256 \end{aligned}$$

Bandwidth:

[
10,000
]

Delay:

[
2,000
]

Add:

[
 $10,000 + 2,000 = 12,000$
]

Multiply:

[
 $12,000 \times 256$
]

[
 $\boxed{3,072,000}$
]

Answer:

EIGRP metric = 3,072,000



HARD EXAMPLE 2 – Successor + Feasible Successor

R2 needs to reach:

192.168.1.0/24

Copy

It has two neighbors:

Neighbor	RD
R3	2,000
R1	1,500

R2's current best FD through R3 is:

FD = 3,000

Copy

Step 1 – Find Successor

Suppose R3 provides the lowest **total path metric**.

Therefore:

R3 = Successor

Copy

Step 2 – Check R1

R1's RD:

1500

 Copy

R2's FD:

3000

 Copy

Check:

[
1500 < 3000
]

☐ True.

Therefore:

R1 = Feasible Successor

 Copy

If R3 fails?

R2 can immediately use:

R1

Copy

without waiting for a full route recalculation.

□ That's why EIGRP converges quickly.



☆ Administrative Distance

Memorize these numbers:

EIGRP route	AD
Internal EIGRP	90
External EIGRP	170
EIGRP Summary	5

Easy memory:



90 → Internal

170 → External

5 → Summary

☆ Important Verification Commands

Check neighbors

```
show ip eigrp neighbors
```

[Copy](#)

Check routing protocols

```
show ip protocols
```

[Copy](#)

Check routing table

```
show ip route
```

[Copy](#)

Look for:

```
D
```

[Copy](#)

because:

D = EIGRP





VERY IMPORTANT EXAM CHEAT SHEET

If you have only a few minutes before the exam, memorize this:

[Copy](#)

EIGRP



Enhanced Interior Gateway Routing Protocol

DUAL



Loop-free + fast convergence

Successor



Best path

Feasible Successor



Backup path

Feasibility Condition



$RD < FD$

Tables



Neighbor → Topology → Routing

Packets



Hello

Update

ACK

Query

Reply

Reliable

↓

Update + Query + Reply

Unreliable

↓

Hello + ACK

IPv4 Multicast

↓

224.0.0.10

Protocol

↓

88

Default Hello/Hold

↓

5 sec / 15 sec

Metric

↓

Bandwidth + Delay

K-values

↓

K1=1 K2=0 K3=1 K4=0

Administrative Distance

↓

Internal = 90

External = 170

Summary = 5

Routing table

↓

D = EIGRP

Router ID priority

↓

Manual → Loopback → Physical

Updates

↓

Partial + Bounded

EIGRP AS

↓

Same AS required for adjacency

The 10 things I'd memorize first

1. EIGRP uses DUAL
2. Successor = best path
3. Feasible Successor = backup path
4. FC: $RD < FD$
5. Neighbor / Topology / Routing tables
6. 224.0.0.10 = EIGRP IPv4 multicast
7. Protocol 88

8. Bandwidth + Delay = default metric

9. AD = 90 / 170 / 5

10. Hello 5 sec / Hold 15 sec

If you know those 10 cold, you've got the **core of EIGRP** nailed down. ☐☐

Absolutely 🤝 This second set is more detailed, but for a **final exam**, you don't need to memorize every sentence. Let's turn it into a **high-yield EIGRP study sheet**: what it means, what to memorize, commands, and **2 easy + 2 hard examples**.



EIGRP FINAL EXAM – HIGH-YIELD NOTES

1. ☆ What is EIGRP?

EIGRP = Enhanced Interior Gateway Routing Protocol

It is an **advanced distance-vector routing protocol** developed by Cisco.

Main ideas:

- Uses **DUAL** to select routes.
- Finds **best paths** and **backup paths**.
- Forms **neighbor adjacencies** with directly connected EIGRP routers.
- Uses **RTP**, not TCP/UDP.
- Sends **partial and bounded updates**.
- Supports **equal-cost and unequal-cost load balancing**.
- Supports IPv4 and IPv6 using **PDMs**.

Remember:



EIGRP = DUAL + Neighbors + RTP + Fast convergence



2. ☆ Three EIGRP Tables

This is **very likely to appear in an exam**.

Table	What it contains
Neighbor Table	Directly connected EIGRP neighbors
Topology Table	All routes learned from EIGRP neighbors
Routing Table	Best routes selected by DUAL

Easy memory:



Neighbor = WHO?

Topology = WHAT routes?

Routing = WHICH route do I use?



3. ☆ PDM – Protocol-Dependent Modules

PDM allows EIGRP to support different network-layer protocols.

Examples:

- IPv4
- IPv6

PDM handles protocol-specific tasks such as:

- Processing packets
- Maintaining tables
- Communicating changes to DUAL
- Calculating protocol-specific metrics

Exam answer:



PDM allows EIGRP to support multiple network-layer protocols such as IPv4 and IPv6.



4. ☆ EIGRP Packet Types

You must memorize these 5.

Packet	Purpose	Reliable?
Hello	Find/maintain neighbors	× No
Update	Send routing information	□ Yes
ACK	Acknowledge reliable packets	× No
Query	Ask neighbors for a route	□ Yes
Reply	Answer Query	□ Yes

Memory

H-U-A-Q-R



Hello → Update → ACK → Query → Reply

Reliable:



Update + Query + Reply

Unreliable:



Hello + ACK

5. ☆ EIGRP Addresses

For IPv4:

EIGRP Protocol Number = 88
EIGRP Multicast = 224.0.0.10

Copy

For IPv6:

Next Header = 88
Multicast = FF02:: A

Copy

☆ Memorize:



EIGRP = 88 + 224.0.0.10



6. ☆ Hello and Hold Timers

Normal Ethernet links

Hello = 5 seconds
Hold = 15 seconds

Copy

Low-speed/NBMA links

Hello = 60 seconds
Hold = 180 seconds

Copy

Exam trick



Normal = 5/15
Low speed = 60/180

If the **hold timer reaches zero**, the neighbor is considered down.



7. Authentication

Authentication checks whether routing updates come from an **authenticated neighbor**.

Important:

Authentication ≠ Encryption

Authentication:



"Who sent this?"

Encryption:



"Can someone read this?"

So EIGRP authentication **does not encrypt routing updates**.



8. ☆ Update Packets

EIGRP does not periodically send the entire routing table like RIP.

Instead it uses:

Partial update

Only the **changed information**.

Bounded update

Only the **routers affected by the change**.

Result:

- ☐ Less bandwidth
- ☐ Less unnecessary traffic
- ☐ Faster convergence



9. ☆ EIGRP AS Number

Example:

```
router eigrp 100
```

Copy

Here:

100 = EIGRP AS number

Copy

Routers forming an EIGRP adjacency must have the **same AS number**.

Example

```
R1 → EIGRP 100  
R2 → EIGRP 100
```

Copy

- ☐ Neighbor can form.

But:

```
R1 → EIGRP 100  
R2 → EIGRP 200
```

Copy

× No adjacency.

Important

The EIGRP AS number is **locally significant**. Don't confuse it with an Internet/BGP AS number.



10. ★ Router ID

Router ID uniquely identifies an EIGRP router.

Priority:

1. Manual Router ID

```
eigrp router-id 1.1.1.1
```

Copy

2. Highest Loopback IP

3. Highest active physical-interface IP

Memorize:



Manual → Loopback → Physical

Router ID cannot be:

```
0.0.0.0  
255.255.255.255
```

Copy

And should be unique.



11. ☆ **network Command**

Example:

```
router eigrp 100  
network 172.16.0.0
```

Copy

This enables EIGRP on matching interfaces.

For a specific /30 :

```
network 192.168.10.8 0.0.0.3
```

[Copy](#)

Wildcard mask

Remember:

```
[  
Wildcard = 255.255.255.255 - Subnet\ Mask  
]
```

For /30 :

```
Subnet mask = 255.255.255.252  
Wildcard   = 0.0.0.3
```

[Copy](#)

12. ☆ Passive Interface

Example:

```
router eigrp 100  
passive-interface g0/0
```

[Copy](#)

A passive interface:

- Does not send EIGRP Hello packets.
- Does not form an EIGRP neighbor adjacency.
- Its network can still be advertised through other active EIGRP interfaces.

Why use it?

- ☐ Security
- ☐ Reduce unnecessary traffic
- ☐ Reduce CPU usage

Very important:



Passive interface does NOT mean the network disappears from EIGRP.



13. ☆ Adjacency Formation

Two routers need compatible EIGRP settings.

Basic process:

```
Hello
↓
Discover neighbor
↓
Form adjacency
```

Copy

↓
Exchange Updates
↓
ACK
↓
Build topology
↓
DUAL selects best paths
↓
Routing table updated

Important requirements

Neighbors need:

- Reachable interfaces
- Same EIGRP AS
- Matching K-values
- Compatible EIGRP configuration



14. ☆ Split Horizon

Split horizon prevents routing loops.

Rule:



A router does not advertise a route back through the same interface from which it learned that route.

Example:

R1 — R2 — Network

Copy

If R2 learns the network from R1, R2 doesn't advertise that same route back to R1 through the same interface.



15. ☆ EIGRP Metric

By default EIGRP uses:

1. Bandwidth

The **slowest bandwidth** on the path.

2. Delay

The **total delay** across the path.

So:

EIGRP default metric = Bandwidth + Delay





16. ☆ K-Values

Default values:

K-value	Parameter	Default
K1	Bandwidth	1
K2	Load	0
K3	Delay	1
K4	Reliability	0
K5	Reliability modifier	0

Most important:

K1 = 1
K3 = 1
K2 = K4 = K5 = 0



Therefore, default EIGRP metric uses:



Bandwidth + Delay

⚠ K-value mismatch

If neighboring routers have different K-values:

× EIGRP adjacency will not form.



17. ☆ Interface Bandwidth

Command:

```
interface g0/0  
bandwidth 10000
```

Copy

The value is in **kbps**.

Important:



`bandwidth` changes the value used by routing calculations. It does **not** physically change the interface speed.



18. Interface Delay Values

Common values:

Interface	Default delay
Ethernet	1000 μs
FastEthernet	100 μs
GigabitEthernet	10 μs
Serial	20,000 μs

Important:

EIGRP uses the **sum of delays** along the path.



19. ☆ EIGRP Metric Formula

For default K-values:

$$\boxed{
 \begin{aligned}
 \text{Metric} = & 256 \\
 & \left(\frac{10,000,000}{\text{Minimum Bandwidth}} \right. \\
 & \quad \left. \bullet \frac{\text{Total Delay}}{10} \right) \\
 & \left. \right)
 \end{aligned}
 }$$

Bandwidth = **kbps**

Delay = **microseconds**

Two things to remember:



Bandwidth → minimum/slowest



Delay → total/sum



● EASY EXAMPLE 1 – Packet Type

Question

Which EIGRP packet is used to discover neighbors?

Answer:

Hello packet ☐

It is:

- Multicast
- Unreliable

- Used to establish/maintain neighbor relationships



● EASY EXAMPLE 2 – AS Number

R1:

```
router eigrp 10
```

[Copy](#)

R2:

```
router eigrp 10
```

[Copy](#)

R3:

```
router eigrp 20
```

[Copy](#)

Which routers can form EIGRP adjacencies, assuming everything else is correct?

Answer:

```
R1 ↔ R2 = ☐
```

```
R1 ↔ R3 = ✕
```

[Copy](#)

R2 ↔ R3 = X

Because:



EIGRP neighbors must use the **same AS number**.



HARD EXAMPLE 1 – Metric Calculation

Suppose a path has:

Minimum bandwidth = 1,000 kbps

Total delay = 20,000 μ s

Copy

Formula:

[
Metric =
256
 $\left(\frac{10,000,000}{1000}\right)$

- $\frac{20,000}{10}$
 $\text{\right)}$
]

Bandwidth:

$$\left[\begin{array}{l} 10,000 \end{array} \right]$$

Delay:

$$\left[\begin{array}{l} 2,000 \end{array} \right]$$

Therefore:

$$\left[\begin{array}{l} \text{Metric} = 256(10,000 + 2,000) \end{array} \right]$$

$$\left[\begin{array}{l} = 256(12,000) \end{array} \right]$$

$$\left[\begin{array}{l} \boxed{3,072,000} \end{array} \right]$$

Answer:

Metric = 3,072,000





HARD EXAMPLE 2 – DUAL + Feasible Successor

R2 wants to reach:

192.168.1.0/24

Copy

It has:

Neighbor	Reported Distance
R3	2,000
R1	1,500

R2's current best path through R3 has:

FD = 3,000

Copy

Step 1 – Successor

Suppose R3 gives the lowest total path metric.

Therefore:

R3 = Successor

 Copy

Step 2 – Check R1

R1 reports:

RD = 1,500

 Copy

R2's FD:

FD = 3,000

 Copy

Feasibility Condition:

[
RD < FD
]

[
1500 < 3000
]

☐ True.

Therefore:

R1 = Feasible Successor

Copy

Step 3 – R3 fails

R2 can immediately change:

Successor: R3

↓

Successor: R1

Copy

No query process is needed because R1 was already a verified loop-free backup.

□ This is the key idea behind EIGRP's fast convergence.



☆ DUAL Terms – MUST KNOW

Term	Simple meaning
DUAL	Chooses loop-free paths
Successor	Best path
FD	Local router's best total metric
RD	Neighbor's advertised metric
Feasible Successor	Loop-free backup

Term	Simple meaning
FC	RD < FD

Super-easy memory:



Successor = BEST ☐

FS = BACKUP ☐

FD = MY BEST DISTANCE

RD = NEIGHBOR'S DISTANCE

FC = RD < FD



20. ☆ Passive vs Active State

Passive

Normal, stable state.

Route is working



No recalculation needed

Copy

Active

The primary route failed and **no Feasible Successor exists**.

The router sends:

Query → Neighbors

Copy

and waits for:

Reply

Copy

Then DUAL calculates a new route.

Remember:



Passive = Peaceful 😊



Active = Problem ☹️



21. ☆ Administrative Distance

Memorize this table:

Route	AD
Connected	0
Static	1
EIGRP Summary	5
Internal EIGRP	90
OSPF	110
RIP	120
External EIGRP	170

Important:

D = Internal EIGRP = 90
D EX = External EIGRP = 170

 Copy



22. ☆ Verification Commands

Check EIGRP neighbors

show ip eigrp neighbors

 Copy

Use it when asking:



"Did my EIGRP neighbors form?"



Check EIGRP configuration

```
show ip protocols
```

Copy

Shows:

- AS number
- Router ID
- K-values
- Passive interfaces
- Administrative distances
- Network configuration
- Routing information sources



Check routing table

```
show ip route
```

[Copy](#)

Look for:

```
D
```

[Copy](#)

Check interface metrics

```
show interfaces
```

[Copy](#)

Useful for:

- Bandwidth
- Delay
- Load
- Reliability



FINAL EXAM – TOP 20 THINGS TO MEMORIZE

1. EIGRP = Enhanced Interior Gateway Routing Protocol
2. EIGRP = Advanced Distance Vector
3. DUAL = Diffusing Update Algorithm
4. DUAL = Loop-free + fast convergence
5. Successor = Best path
6. Feasible Successor = Backup path
7. Feasibility Condition = $RD < FD$
8. Neighbor Table = Direct neighbors
9. Topology Table = All learned routes
10. Routing Table = Best routes
11. EIGRP Protocol = 88
12. IPv4 multicast = 224.0.0.10
13. IPv6 multicast = FF02:: A
14. Hello = Neighbor discovery
15. Update = Routing information
16. ACK = Acknowledgment
17. Query = Ask for route

18. Reply = Answer Query

19. Default Hello/Hold = 5/15 seconds

20. Default metric = Bandwidth + Delay

And these numbers!

K1 = 1

K2 = 0

K3 = 1

K4 = 0

K5 = 0

AD:

Summary = 5

Internal = 90

External = 170

Protocol:

EIGRP = 88

IPv4 Multicast:

224.0.0.10

Timers:

5 / 15 seconds

60 / 180 seconds on low-speed NBMA

 Copy

One-line EIGRP story

Hello finds the neighbor → **Update** shares routes → **RTP/ACK** makes delivery reliable → **Topology table** stores possibilities → **DUAL** chooses the **Successor** → **Feasible Successor** provides backup → **Routing table** installs the best path. □

That sequence is **very worth memorizing for the final**.